Anika Thapar

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Education

Purdue University — College of Sciences West Lafayette, IN
B.S. in Computer Science, Minor in Mathematics, Certificate in Entrepreneurship & Innovation *May 2028*

- **GPA:** 3.77/4.0
- **Courses:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Software Testing

Work Experience

Software Engineering Intern Oct 2025 – Aug 2026
Flovision *Remote*

- Engineered a DVR buffer system for live HLS streams using **GCP** (Cloud Run, Cloud Storage, Valkey, BigQuery) synchronized with custom **Flutter** client engine, enabling customers to rewind up to 30 minutes of live playback.
- Integrated **L2 caching** layer using Valkey and optimized backend queries, containerized with **Docker** and deployed via **AWS ECS and ECR**, reducing analytics page load time from 3.2s to **under 50ms.**
- Optimized a computer vision data pipeline by managing **AWS S3 buckets** containing 1,000+ hours of raw facility video data, benchmarking multiple OCR engines to achieve **99.5%** text detection accuracy.

Fullstack Software Engineer Feb 2025 – May 2025
Continuity Pharma *West Lafayette, IN*

- Developed persistent **Python-based** relay client on a Raspberry Pi connecting to an EC2-hosted backend, implementing exponential backoff and heartbeat monitoring, reducing manual restarts by **95%**.
- Optimized backend data pipelines to support partial system operations, maintaining uninterrupted data flow.

Software Engineer Intern May 2025 – Aug 2025
Sierra Ventures *San Mateo, CA*

- Architected an automated data scraping pipeline using BeautifulSoup to extract **1,500+** startup profiles, reducing manual data entry by 15 hours/week and enabling real-time due diligence processing for the investment team.
- Created Python workflows with **OpenAI API**, applying prompt engineering to enrich due diligence and sourcing.

Backend Engineer Intern Jun 2024 – Aug 2024
Arrcus *San Jose, CA*

- Developed Python script using pexpect and **multi-threading** to detect core dumps, improved detection by **5x**.
- Implemented 4 Jinja2 templates to streamline configuration testing across multiple networking protocols.

Projects

Refactorika | *CalHacks Winner* Jun 2026

- Developed Refactorika, an open-source **Model Context Protocol** (MCP) server that empowers LLMs with deterministic, verified structural-refactoring capabilities over complex multi-language codebases.
- Designed architecture that translates code-bases to AST-node transformations, reducing token usage by **40%**.

Barter | *CS 307* Jan 2026 – May 2026

- Led a team of 4 to build a full-stack marketplace using **Node.js** and **Express**, enabling seamless item trading.
- Implemented **PostgreSQL** triggers to invalidate JWT signatures across multiple browsers simultaneously.
- Leveraged **Supabase** to manage backend services, including automated login and user preference workflows.

Leadership

CS Undergraduate Student Board Feb 2025 – Present

- Serve as a liaison between 3,500+ students and faculty to identify department concerns and spearhead initiatives.
- Founded and scaled the Panels initiative and Social Wellness initiative, leading 15 students in organizing events including student forums and career workshops, enhancing student's academic and social well-being.

Kappa Theta Pi Purdue Feb 2026 – Present

- Founding member and Vice President of Purdue's largest professional technology fraternity, guiding students in career preparation through hosting development workshops, hackathons, speaker series, and mock interviews.
- Averages 15,000+ views per post on Instagram and LinkedIn, attracting 150+ students to callout events.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML/CSS
Frameworks: Node.js, React, Next.js, Flutter/Dart, Express, TensorFlow, OpenCV
Developer Tools & Cloud: Git, Linux, Docker, AWS (ECS/ECR/EC2/S3/SQS), Google Cloud (BigQuery/Memorystore/Cloud Run/Cloud Storage), Redis, Jenkins, MongoDB, Supabase, PostgreSQL